

Sanaa Anjum

Mathematics Lecturer/ Researcher (Mathematics & Data Science)/ Advanced Mathematics LLM Trainer
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SUMMARY

A devoted Mathematics Lecturer and Researcher with extensive experience in teaching, data science, and academic research, currently pursuing a PhD in Mathematics (expected to be completed by Fall 2025), specializing in algebra, analysis, and their applications in machine learning and computational biology. Over a decade of experience in university-level instruction, proficient in developing student-centred curricula, promoting critical thinking, and mentoring students in advanced mathematical concepts. Proficient in algebraic structures, machine learning applications, and computational modelling using Python and LaTeX. Design and refine advanced mathematical problems and proofs using LaTeX, providing expert feedback on AI-generated solutions and collaborating with researchers to improve model performance and dataset quality. Skilled in collaborating with faculty and academic stakeholders to enhance learning experiences and community engagement. Determined to support creativity in mathematics and data science through continuous professional development and interdisciplinary cooperation.

PROFESSIONAL EXPERIENCE

Lecturer

University of Wah, Department of Mathematics

Mar '12 — Present

Wah Cantt, Pakistan

Played a key role in enhancing student learning through innovative teaching methodologies, curriculum development, and mentorship, fostering critical thinking and career growth. Committed to continuous professional development, staying updated with advancements in education and research to drive institutional excellence.

- Developed high-impact learning content in Algebra and Analysis, incorporating innovative teaching methodologies and diverse learning modalities to improve student engagement and comprehension.
- Contributed to the departmental curriculum revision committee, aligning undergraduate courses with HEC standards, ensuring a modern and research-driven educational framework.
- Provided mentorship and academic advising, fostering critical thinking, problem-solving abilities, and career exploration, helping students excel academically and professionally.
- Coordinated with research teams, leading to meaningful contributions in mathematical and computational research, strengthening the university's academic reputation.
- Actively participated in training programs and professional development opportunities, staying updated with cutting-edge advancements in education and research to enhance institutional excellence.

Clinical Data Scientist (Part-time)

Continuum Research Center

Oct '22 — Present

Wah Cantt, Pakistan

Optimized data workflows, developed predictive models using machine learning and statistical techniques, enhancing clinical outcome forecasting and research insights.

- Optimized clinical data workflows by gathering, cleaning, and integrating data from multiple sources, enabling comprehensive analysis for improved decision-making.
- Developed predictive models using machine learning and statistical techniques, enhancing the accuracy of clinical outcome forecasting and supporting evidence-based decision-making.
- Designed clinical studies and developed protocols, incorporating advanced data visualization to effectively communicate findings to diverse stakeholders.
- Prepared detailed reports and presentations, translating complex clinical data into actionable insights for researchers, clinicians, and decision-makers.
- Fostered cross-functional collaboration by working closely with clinical teams, biostatisticians, and data managers, ensuring the successful execution of clinical studies and adherence to research standards.

Advanced Mathematics LLM Trainer & Research Analyst

Turing Enterprises Inc

Aug '25 — Present

Remote

Responsible for designing and refining advanced mathematical problems, solutions, and proofs using LaTeX, ensuring accuracy across pure and applied mathematics.

- Crafting, testing, and refining high-quality prompts to guide AI in generating accurate mathematical reasoning and proofs.
- Evaluate and provide rigorous feedback on AI-generated outputs to ensure accuracy, clarity, and academic relevance.
- Writing, editing, and formatting mathematical manuscripts and technical reports using LaTeX.
- Collaborated with interdisciplinary global teams of AI developers and researchers to improve model quality.

EDUCATION

Ph. D in Mathematics, International Islamic University (GPA: 3.75)

Present

Islamabad, Pakistan

- Title of PhD Thesis: A Study on Homomorphisms of Generalized Algebraic Structures of DNA Sequences and their Applications

M. Phil. in Mathematics, Quaid-i-Azam University (GPA: 4.2, Fellowship in 1st and 2nd Semesters)

Islamabad, Pakistan

- Title of M. Phil. Thesis: Characterizations of Hemirings by the Properties of their Interval Valued T fuzzy Ideals

CERTIFICATIONS

Good Clinical Practice, NIDA Clinical Trials Network	Apr '23
Machine Learning Algorithms, Great Learning Academy	Aug '23
Python for Data Science and AI, Coursera, authorized by IBM	Nov '23
Python for Genomic Data Science, Coursera, authorized by Johns Hopkins University	May '24
Machine Learning with Python, Coursera, authorized by IBM	Jun '24
English for Common Interactions in the Workplace: Basic Level, Coursera, authorized by Pontificia Universidad Cat3lica de Chile	Jun '25
Machine Learning Basics, Coursera, authorized by Sungkyunkwan University	Jun '25
NVIDIA: Fundamentals of Machine Learning, Coursera, authorized by Whizlabs	Jun '25

SELECT ACHIEVEMENTS

Authored papers in peer-reviewed, high-impact journals gained expertise in developing numerical mapping methods using algebraic structures, genomic data analysis, mathematical modelling, and machine learning algorithms for mutation detection, and interdisciplinary research in computational biology.

PROJECTS

Application of amplitude-modulated transcranial pulsed random noise stimulation (am-tPRNS) in Parkinson’s Disease, Continuum Research Centre	Present Wah Cantt
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SKILLS

Communication Skills Communication, Interdisciplinary Collaboration, Feedback Delivery
Analytical Skills Detailed Oriented, Analytical & Logical, Problem-Solving, Critical Evaluation
Technical Skills Machine Learning, Data Science, LaTeX Expertise, Python
Leadership Skills Leadership & guidance, Team Collaboration
Personal Attributes Fast learner, Time management

PUBLICATIONS

Factors Influencing Breast Cancer Knowledge and Attitudes of Women in Abbottabad, Pakistan.
Medical Science Monitor
Muhammad Arfat Yameen, Sanaa Anjum, Warda Khan, Atif Ali, Hira Khan, Zhongkun Zhang, Naji Alqahtani, Sana Samreen, 2025, 3 1 : e947608. IF: 2.2

Novel DNA Numerical Mapping via Pseudo-Metric Free Semigroups for Machine Learning-Enabled Sequence Comparison Tasks.
Fourrages journal
Sanaa Anjum, Sajida Kousar, Nasreen Kausar, 2024, 260 (10); 22-39. IF: 0.202

Digitizing DNA Sequences Using Multiset-Based Nucleotide Frequencies for Machine Learning-Based Mutation Detection.
Decision Making Applications in Management and Engineering
Sanaa Anjum, Sajida Kousar, Nasreen Kausar, Nezir Aydin, Oludolapo Akanni Olanrewaju, Bongumenzi Mncwango, 2024, 7; 516-529. Scopus, Q1.

On the generalization of Hermite-Hadamard type inequalities for E-convex function via functional integrals.
Heliyon
Muhammad Sadaqat Talha, Tasawar Abbas, Zhixiong Li, Amor Rebey, Affaq Ahmad, Sanaa Anjum, 2024, 10 (10); e3 104. IF:4.0

Numerical simulations for radiated bioconvection flow of nanop with viscous dissipation and exponential heat source.
Journal of the Indian Chemical Society
Kaouter Ghachem, Bilal Ahmad, Skeena Noor, Tasawar Abbas, Sami Ullah Khan, Sanaa Anjum, Norah Alwadai, Lioua Kolsi, 2023, 100(1). IF: 0.284

Trends in Prescribing Patterns and Drug Related Problems of Kidney Disease Patients.
Journal of Pakistan Medical Association
Muhammad Arfat Yameen, Mubashra Tafseer, Warda Khan, Sanaa Anjum, Raza-E-Mustafa, Ossam Chohan, 2021, 71(11); 2629-2636. IF: 0.573

Application of interval t-norm fuzzy ideals of hemirings with intervals valued characteristics function.
Annals of Fuzzy Mathematics and Informatics
Sanaa Anjum, Bilal Ahmed, Tasawar Abbas, 2020, 20(1); 15-27. Non-ISI

REFERENCES

- Dr. Muhammad Shabir (Professor) Quaid-i-Azam University, Islamabad, Pakistan,
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- Dr. Muhammad Arfat (Associate Professor) COMSATS University Islamabad, Abbottabad Campus, Abbottabad, Pakistan
arfatyameen@gmail.com (+92 333 5760361)
- Dr. Bilal Ahmad (Assistant Professor) University of Wah, Wah Cantt, Pakistan
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